

# REZA ALAEDDINI

## Forensic Biologist

@ ralaeddini@ysu.edu    +1 (330) 941-3572    in reza-alaeddini-84151142    Ward Beecher Hall 3041    Ohio, USA

## PROFILE

PhD in Molecular Biology, plus 20 years of concrete knowledge and experience in the pursuit and practice of basic and applied Forensic Science.

## VOCATIONAL EXPERIENCE

### Youngstown State University, OH

#### Assistant Professor of Forensic Science

- Teaching the following courses:
  - Forensic DNA Analysis and interpretation
  - Forensic Pathology

### Columbia College, MO

#### Visiting Assistant Professor of Forensic Science

- Taught the following courses:
  - Forensic DNA Analysis and interpretation
  - Forensic Pathology
  - Professional issues in Forensic Science
  - Physical and Chemical Methods in Forensic Science
  - Senior Seminar in Forensic Science and presentations

### UNT Dallas & Texas Wesleyan University, TX

#### Adjunct Instructor

- Taught the following courses:
  - Forensic Biology
  - Genetics
  - Theoretical and Laboratory Biology
  - Laboratory Immunology

### Tarrant County Medical Examiner's Office, TX

#### Senior Forensic Scientist

- Validated and implemented the framework for statistical interpretation of DNA mixture profiles on STRmix software.

### Forensic Biology Laboratory

#### Senior Forensic Scientist

- Worked as a case worker in human DNA profiling.
- Supervised the implementation and validation of DNA Data basing software.
- Designed a PCR based DNA quantification methodology of mitochondrial DNA by using LUX fluorescent primers on ABI 7500 Real-time PCR machine.

## WORK EXPERIENCE

### Youngstown State University

#### Assistant Professor of Forensic Science

Aug 2025 – Ongoing

### Columbia College

#### Visiting Assistant Professor of Forensic Science

Aug 2022 – May 2024

### UNT Dallas & Texas Wesleyan University

#### Adjunct Instructor

Aug 2019 – June 2022

### Tarrant County Medical Examiner's Office

#### Senior Forensic Biologist

2016 – 2018

### Forensic Biology Laboratory, LMO

#### Senior Forensic Biologist

2010 – 2015

## MEMBERSHIPS

- American Academy of Forensic Sciences
- International Assoc. for Identification

## SKILLS

- Probabilistic DNA genotyping
- Ancient DNA purification protocols
- Kinship analysis
- Statistical interpretation of DNA profiles
- Mt DNA phylogenetics
- Quantitative PCR
- ESEM microscopy
- STRmix Software
- R program Basic Packages
- TeX, LaTeX typesetting

## PUBLICATIONS

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### Journal Articles

- Alaeddini, R., M. Yang, and B. Puza (2020). "Bayesian regression analysis of stutter in DNA mixtures". In: *Communications in Statistics - Theory and Methods* 4 (12), pp. 1–15.
  - Mehrafshan, N., M. Ahmadi, and R. Alaeddini (2015). "Development of a Pentanucleotide STR Marker for Human Identity Testing". In: *Journal of Biology and Today's World* 4 (12), pp. 204–208.
  - Alaeddini, R. (2012). "Forensic implications of PCR inhibition—A review". In: *Forensic Sci Int Genet* 6 (3), pp. 297–305.
  - Alaeddini, R. and M. Ahmadi (2011). "Inhibitory effects of bone extracellular matrix components on PCR". in: *Clinical Biochemistry Supplement* 44 (13), p. 259.
  - Alaeddini, R., M. Ahmadi, et al. (2011). "Semi-quantitative PCR Analysis of DNA Degradation". In: *Australian Journal of Forensic Sciences* 43 (1), pp. 53–64.
  - Alaeddini, R., S. J. Walsh, and A. Abbas (2010). "Forensic implications of genetic analysis from degraded DNA". in: *Forensic Sci Int Genet* 4 (3), pp. 148–157.
  - Alaeddini, R., A. Abbas, and S. J. Walsh (2009). "Molecular studies of time and environment dependent effects on bone DNA survival". In: *Australian Journal of Forensic Sciences* 42 (3), pp. 211–220.
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### Conference Proceedings

- Alaeddini, R. (2017). "Introduction to MCMC methods for resolving DNA mixtures". In: *Southwestern Association of Forensic Scientists*. Fort Worth, Texas.
- Alaeddini, R. and A. Abbas (2012). "Development of a mitochondrial DNA based qPCR assay for the assessment of DNA degradation in forensic samples". In: *The 6th European Academy of Forensic Science Conference*. The Hague, Netherlands.
- Alaeddini, R. and M. Ahmadi (2012). "Micro-ATR infrared spectroscopic screening of bone samples for the assessment of DNA survival". In: *The 6th European Academy of Forensic Science Conference*. The Hague, Netherlands.
- Alaeddini, Reza (2012). "Water purification essentials in forensic genetics laboratory". In: *The 6th European Academy of Forensic Science Conference*. The Hague, Netherlands.
- Alaeddini, R., S. J. Walsh, and A. Abbas (2008). "Complications of PCR based genotyping in forensic context". In: *The 19th International Symposium on the Forensic Sciences*. Melbourne, Australia.
- Alaeddini, R., R. Zhong, et al. (2008). "Applications of infrared spectroscopic analysis in forensic bone samples". In: *The 19th International Symposium on the Forensic Sciences*. Melbourne, Australia.

## EDUCATION

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### PhD Molecular Biology

University of Sydney, Australia

 2004 – 2009

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### MS Forensic Medicine

Tehran University, Iran

 1999 – 2003

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### Bachelor of Medicine

Melli University, Iran

 1987 – 1995

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## AWARDS

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- NIFS award for the best review paper of the year, Australia 2010
- Certified Medical Laboratory Technical Officer, Australia 2007
- IPRS scholarship covering PhD tuition and stipend, Australia 2004
- Research Scholarship on bone DNA extraction and amplification at Flinders University, Australia 2001